QUANTUM COMPUTING INC

Quantum-Accelerated Optimization for Network Planning

Scheduling, routing, and frequency assignment on Dirac-3 hardware

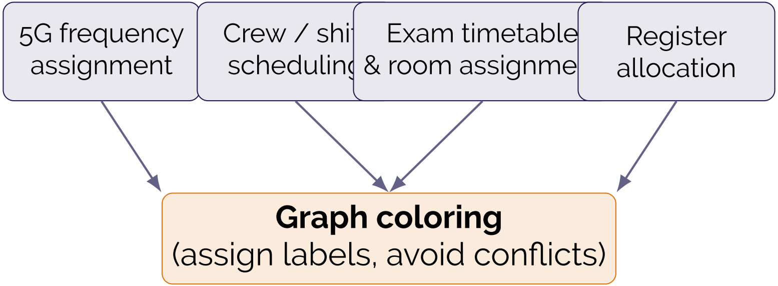
QCi Solutions

Quantum Computing Inc

May 12, 2026

Outline

One problem, many faces



Every problem above reduces to minimum vertex coloring: assign one of k labels per item so conflicting items differ. We want k as small as possible.

Why this matters at scale

The economic stake

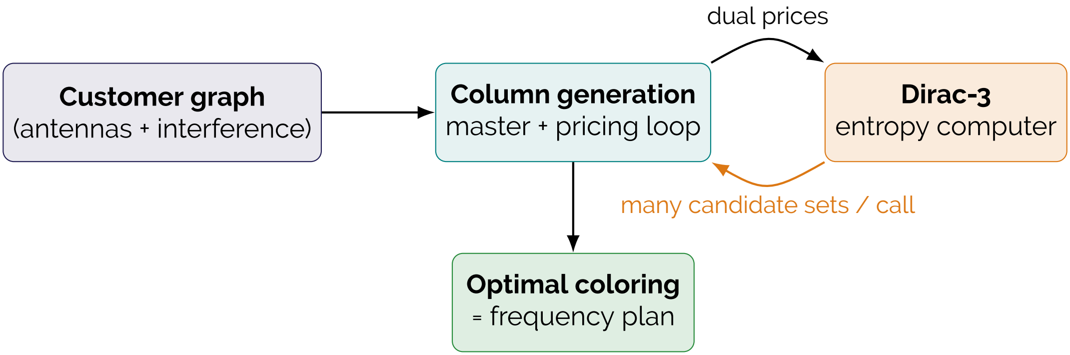
A 5G operator with 5000 antennas across a metro area must minimise spectrum use. Each saved frequency translates to capacity that can be sold to subscribers.

Today's tooling:

- Greedy heuristics: fast, off by 20–40% on dense networks.
- Exact solvers (CPLEX, Gurobi): exact, but don't scale past a few dozen nodes on dense graphs.
- Custom column generation: scales but slow on the inner "best next column" subproblem.

QCi's approach: accelerate the inner subproblem on Dirac-3.

The QCi pipeline



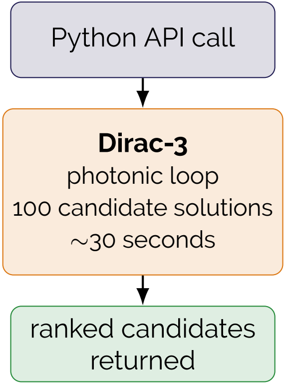
- No quantum compilation by the customer; the device is wrapped behind a Python API.
- One device call returns dozens of candidate assignments in parallel — the wedge that beats classical CG.

Dirac-3 in one slide

Dirac-3 is QCi's **entropy computer**: a photonic device that *natively* optimises a continuous quadratic program in a single shot.

- Each call returns up to 100 high-quality solution candidates.
- No quantum-circuit compilation, no qubit-count bottleneck.
- Cloud-API access today; on-prem deployment available.

The hard combinatorial subproblem inside column generation maps directly onto Dirac's native interface.



Head-to-head: same optimum, fewer iterations

Method	Frequencies used	Wall time	Iterations
Greedy (DSATUR)	8	< 0.01 s	—
Direct optimisation	6	0.04 s	—
Classical CG	6	0.02 s	7
QCi Quantum CG	6	50 s	**4**

Synthetic 12-antenna interference network. All methods that reach 6 match the proven optimum. QCi Quantum CG converges in roughly half the iterations of classical CG — the gap widens on larger, denser instances where each iteration costs more.

Where the wedge widens

Benchmark suite (ER random graphs)

- $n = 100, p = 0.5$: Quantum CG returns 2–4$\times$ more candidate solutions per call; converges in half the iterations.
- $n = 100, p = 0.7$: classical solvers degrade quickly; QCG reaches the same chromatic number with a richer column pool.
- $n > 100$: classical exact methods time out; CG remains the only tractable path, and QCG accelerates the inner loop.

Per-call latency on Dirac-3 cloud is currently $\sim$30–90 s. Wall-clock advantage materialises when the classical alternative also takes minutes per iteration.

What we deliver

- A Python toolkit that drops into existing planning workflows; the customer writes a NetworkX graph, the toolkit returns a coloring.
- Three backends interchangeable behind one API:
  - Cloud Dirac-3 (production today)
  - On-prem Dirac-3 (pilot)
  - Classical CG fallback (always available)
- Reproducibility built-in: recorded device calls can be replayed offline, so colleagues review the same results without burning device time.

Try it

This bundle

- 6 Jupyter notebooks (theory $\rightarrow$ end-to-end pipeline).
- 3 slide decks (this one, an end-to-end tutorial, a technical deep dive).
- All examples run offline on bundled data; live device mode is one toggle away.

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